Shared Biomarker Effects, Prognostic Biomarkers and the Symmetry Screen: Five Issues from the Hybrid Stress Test

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Purpose. The stress test set out in `docs/46` was run on 2026-10-06 and 07: 39 cells, 26,500 simulated trials and 11 analyses, with 27 failed fits in 291,500. Its results turn on five issues. This paper explains each from first principles and gives the evidence for it:

1. what it means for an analysis to *share* the biomarker effect with baseline;
2. why a prognostic biomarker breaks every analysis that does so;
3. what *freeing* the baseline biomarker effect does;
4. the symmetry screen of the working protocol, and why it failed;
5. the untested combination of the recommended analysis with phase-specific variances.

The full results report, against the working protocol, is reserved for `docs/47`.

Update (2026-10-07). After this paper was drafted, two baseline-severity cells (40 and 41) and three post hoc analyses (G1, G2, F1A) were run on the same trials, bringing the study to 41 cells, 28,000 trials and 14 analyses (`docs/46`, Amendment 2). The figures below for the original 11 analyses and 39 cells are unchanged; the new results are added in Sections 2, 3 and 5.

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Notation index and glossary

Notation follows `analysis/report/NOTATION.md`, `docs/45` and `docs/46`. Symbols marked *local* are defined here.

Table 1: Notation index and glossary of symbols, meanings and status

Symbol	Meaning	Status
Y_{it}	symptom score (code `Sx`); $t = 0$ is baseline	canonical
B_i	biomarker (blood pressure); centered in every analysis (code `bmc`)	canonical
$D_{bc,it}$	exposure-decayed drug indicator (code `Dbc`), 0 at baseline	canonical
β_B	biomarker main effect in the analysis model	local (`docs/41`)
$\beta_{bm:D}$	biomarker-by-drug interaction, the estimand	canonical
s_0, s_{off}, s_{on}	true slope of Y on B at baseline, at post-baseline off-drug visits, at on-drug visits	local
s_{prog}	prognostic slope: the slope of Y on B that does not depend on drug	local
f	the share of s_{prog} that a shared β_B absorbs	local
c_{bm}	biomarker-BR coupling (the predictive effect)	canonical
$c_{bm,BL}$	biomarker-baseline correlation (baseline-severity prognostic coupling)	local (`docs/46`)
μ_B, B^*	mean of the biomarker; the biomarker reflected about it, $2\mu_B - B$	local
π	power	canonical

Analyses (`docs/46`, Section 4). All of them use a random intercept, continuous-time AR(1) residuals (`corCAR1`) and CR2 standard errors unless stated, with baseline as a response row, linear time, and $D_{bc,it}$ at an assumed half-life of 0.5 week.

- **S** (the strawman): `Sx ~ bmc + t + Dbc + bmc:Dbc`.
- **ES**: S plus `bmc:bl0`, where `bl0` is 1 at baseline.
- **F1**: S plus `De + bmc:De`, where `De` is the expectancy: 0 at baseline, 1 open label, 0.5 blinded.
- **F2**: S plus `bmc:t`.
- **C1**: S with residual variance by phase, `varIdent(~ 1 | phase)`, for the phases baseline, open label and blinded. **C2** is C1 with model-based inference.
- **A**: unstructured covariance with Kenward-Roger. **EA** is A plus `bmc:bl0`.
- **DS** and **DA**: S and A with the half-life chosen by AIC.

- **R**: the published analysis.
- **G1**, **G2** (post hoc): F1 with residual variance by phase, with CR2 and with model-based inference. **F1A** (post hoc): F1 with the half-life chosen by AIC.

Glossary.

- **Predictive biomarker.** Changes the effect of the drug.
- **Prognostic biomarker.** Associated with the outcome whatever the treatment, for example through the natural course or the placebo response.
- **Size.** The false-positive rate of the interaction test; 0.05 nominal.
- **Size-adjusted power.** Power when each analysis rejects at its own 5% null quantile, so analyses are compared at equal size.

1. The shared biomarker effect

How strongly does the symptom score move with blood pressure? Asked separately in three groups of rows, the question gives three slopes:

Table 2: True slope of Y on B in three groups of rows

Rows	True slope of Y on B
baseline (week 0, before any treatment)	s_0
post-baseline visits off drug	s_{off}
post-baseline visits on drug	s_{on}

The estimand is the extra biomarker effect on drug, $\beta_{bm:D} = s_{\text{on}} - s_{\text{off}}$.

The analyses that keep baseline as an outcome row coded off drug (S, A, C1, C2, DS, DA, F2, R) have only two slope parameters:

- β_B (`bmc`) is the slope in *every* row coded off drug: the baseline row and the post-baseline off-drug visits together;
- $\beta_{bm:D}$ (`bmc:Dbc`) is the additional slope on drug.

So the model imposes $s_0 = s_{\text{off}}$: a single β_B describes baseline and the later off-drug visits. This is what *sharing the biomarker effect with baseline* means. The condition that makes it correct, $s_0 = s_{\text{off}}$, is the constancy assumption.

When the assumption holds, for a purely predictive biomarker, $s_0 = s_{\text{off}} = 0$ and only s_{on} differs from zero. The constraint is then correct, and the baseline row adds a clean off-drug observation that sharpens β_B, and through it $\beta_{bm:D}$. That free precision is why the shared-slope analyses were the most powerful under AR(1) data (verified):

Table 3: Size-adjusted power of shared and freed analyses, AR(1) construct

Size-adjusted power, AR(1) construct, $c_{bm} = 0.30$	
S (shared)	0.514
ES (freed)	0.364

The mechanism, the extra information the baseline row contributes to β_B, was isolated in `docs/45`, Section 6, by freeing that one term.

2. Why a prognostic biomarker breaks it

Suppose blood pressure predicts the natural course of symptoms. For example, participants with higher pressure improve more over time, whether or not they take the drug. With no true interaction:

- $s_0 = 0$: no course has happened yet at baseline;
- $s_{\text{off}} = s_{\text{on}} = s_{\text{prog}}$;
- the true interaction is $s_{\text{on}} - s_{\text{off}} = 0$.

The shared-slope model must fit one β_B to the baseline rows (true slope 0) and to the off-drug visits (true slope s_{prog}). The fit settles between them, $\hat{\beta}_B \approx f\, s_{\text{prog}}$ with $0 < f < 1$. The on-drug rows have slope s_{prog}, and the only parameter left to account for the gap is the interaction:

$$\hat{\beta}_{bm:D} \approx s_{\text{prog}} - f\, s_{\text{prog}} = (1 - f)\, s_{\text{prog}} \neq 0.$$

The model therefore reports an interaction that does not exist. The error is a fixed bias, not noise, so the false-positive rate grows with the sample size.

How much leaks depends on how much weight the baseline row receives when β_B is estimated. The more weight it gets, the smaller f and the larger the leak.

Table 4: Baseline weight and null rejection under prognostic biomarker cells (1,000 simulated trials per cell)

Analysis	Weight on baseline	Null rejection: TV, constant	TV, growing	PB
S	moderate	0.099	0.049	0.100
DS (S with AIC)	moderate	0.115	0.058	0.112
C1 (S, phase variances)	larger	0.382	0.173	0.205

Analysis	Weight on baseline	Null rejection: TV, constant	TV, growing	PB
A (unstructured, KR)	heavy	0.416	0.195	0.222
R (published)	moderate	0.107	0.053	0.159

TV: blood pressure correlated 0.3 with the natural-course component, at every visit (constant) or growing along its curve. PB: correlated 0.3 with the placebo component. The rates are verified; the weighting account is inferred.

Baseline severity does not break it (verified, cell 40, added 2026-10-07). When blood pressure is correlated 0.3 with the baseline level, its association with Y is the same at every visit, so $s_0 = s_{\text{off}}$ and the constraint is correct. Every analysis is then within the cell limit: S 0.050, DS 0.055, C1 0.054, A 0.051 and R 0.075 (the limit itself).

- **The unstructured model fails worst.** It weights the baseline row most heavily, because baseline is the off-drug observation least contaminated by treatment.
- **Phase-specific variances make the leak worse.** The baseline variance is the smallest, so the phase-variance model up-weights the baseline row.
- **A placebo-prognostic biomarker** breaks the constraint the same way: the placebo response is 0 at baseline and nonzero afterwards, so again $s_0 \neq s_{\text{off}}$.
- **A natural-course association that grows over time** leaks least into S, because it is small early, which is also where most on-drug visits are.

3. Freeing the baseline biomarker effect

The direct remedy gives baseline its own slope:

```
Sx ~ bmc + t + Dbc + bmc:Dbc + bmc:bl0
```

Table 5: Slope of the biomarker in the model for each group of rows

Rows	Slope in the model
baseline	β_B + coefficient of `bmc:bl0` (whatever the data say)
post-baseline off drug	β_B
on drug	$\beta_B + \beta_{bm:D}$

The interaction is now identified from post-baseline visits alone, so a difference between the baseline slope and the post-baseline slope no longer leaks into it.

This is ES. (Note 2026-10-08: ES is not immune to every prognostic pathway. A slope that scales with expectancy, as in the placebo-prognostic case, is larger at the open-label visits, which are all on drug, and leaks into $\beta_{bm:D}$ by a factor of 0.14 to 0.24 of s_{prog} depending on the working covariance (verified, published construct; `analysis/report/compendium-mathematics.tex`). The stress-test rejection rate of 0.049 in the PB cell is consistent with a small leak.) It is equivalent to analyzing baseline as a covariate (`docs/41`, Section 7, confirmed in `docs/45`, Section 6).

F1 frees baseline implicitly. It adds the expectancy and its interaction with the biomarker, `De + bmc:De`. Because the expectancy is 0 at baseline and positive afterwards, the biomarker slope at baseline is no longer tied to the post-baseline slope. In addition, the slope may scale with the expectancy, which matches the placebo-prognostic case directly.

Table 6: Null rejection of analyses that free the baseline biomarker effect (verified)

Analysis	Pooled null, 11 standard cells	TV, constant	TV, growing	PB
ES	0.048	0.049	0.063	0.049
EA	0.052	0.049	0.057	0.050
F1	0.051	0.047	0.046	0.040
F2 (`bmc:t` only)	0.049	0.090	0.048	0.098

In the baseline-severity null cell (added 2026-10-07) these rates are ES 0.050, EA 0.066, F1 0.040 and F2 0.048 (verified).

F2 shows that modeling the time course alone is not enough. It repairs only the growing natural-course case, because it does not free the baseline slope.

The cost. When the biomarker is purely predictive, freeing the slope throws away the information the baseline row contributed. Under AR(1) data this costs about a third of the power (0.364 against 0.514 above). Under compound symmetry the participant effect already supplies that information, and the two analyses part ways (verified): ES gains power over S (0.662 against 0.554, a difference of 0.108), while F1 still loses a little (0.522, a difference of 0.032). The mechanism behind ES's gain is not derived in any document.

Table 7: Size-adjusted power of S, ES and F1 in the CS construct

Size-adjusted power	S	ES	F1
CS construct, $c_{bm} = 0.25$	0.554	0.662	0.522

Choosing between ES and F1. Both pass every size screen. They differ in power by condition (verified):

Table 8: Size-adjusted power of ES and F1 by cell, with maximin shortfall of each relative to the better of the two

Size-adjusted power, alternative cells	ES	F1
maximin shortfall, non-prognostic cells	0.046	0.140
TV, constant prognosis, $c_{bm} = 0.30$	0.370	0.482
TV, growing prognosis, $c_{bm} = 0.30$	0.192	0.378
PB prognosis, $c_{bm} = 0.30$	0.498	0.446
maximin shortfall, all cells	0.186	0.140

The shortfalls here compare ES and F1 with each other only. Measured against the best nominal survivor in each cell, as in the decision rule, they are larger: on the non-prognostic cells F1 0.242 and ES 0.278, so F1 ranks ahead of ES (`docs/46`, Amendments 1 and 2; verified, `decision-maximin-ds-nosym-all14.csv`).

- When the biomarker is purely predictive, ES is slightly better head to head (mean size-adjusted power 0.352 against 0.341 over the 22 non-prognostic cells), although its worst shortfall from the best survivor is larger.
- When it is prognostic, F1 keeps its power across all three forms; ES loses most of it when the association grows over time.

The ranking therefore depends on whether the prognostic alternative cells count. The working protocol does not settle this; `docs/47` will report both versions. `docs/46` recommends F1 because it is robust in size and in power.

4. The symmetry screen

What the protocol specified. One component of sensitivity (`docs/46`, Section 3) required that power at $-c_{bm}$ be within 2 Monte Carlo SE of power at $+c_{bm}$. It was tested at ± 0.30 in the AR(1) construct and ± 0.20 in the compound-symmetry construct. An analysis failing either test could not be selected.

Why power is exactly symmetric (derived). In these cells the biomarker is correlated only with the drug-response component. Let $B^* = 2\mu_B - B$, the biomarker reflected about its mean. B^* has the same mean and variance as B, and its covariance with every other variable has the opposite sign. So the joint distribution of $(B^*, \text{everything else})$ under $+c_{bm}$ is exactly that of $(B, \text{everything else})$ under $-c_{bm}$.

Every analysis uses the centered biomarker linearly, and negating that column negates $\widehat{\beta}_B$ and $\widehat{\beta}_{bm:D}$ while leaving every fitted value, variance estimate, standard error, degree of freedom and p-value unchanged. The two-sided rejection rate at $-c_{bm}$ therefore equals the rate at $+c_{bm}$. The size-adjusted rates are equal too, because both cells use the same null quantile.

Any observed difference is Monte Carlo error.

What happened. The $-c_{bm}$ and $+c_{bm}$ cells are separate, independently seeded simulations. The analyses within a cell are fitted to the same trials and are therefore strongly correlated.

- In both constructs, power came out higher at $+c_{bm}$ for every analysis, by 0.004 to 0.084 (verified).
- At the threshold, about 0.06, 9 of the 22 tests failed: S, DS and F2 in the AR(1) construct; A, C1, EA, ES and R in the compound-symmetry construct.
- The screen thereby removed the strawman and ES for no real reason.

Rerun with fresh seeds (`13-symmetry-check.R`, 1,000 trials per cell, verified):

Table 9: Power at $-c_{bm}$ and $+c_{bm}$ in the symmetry rerun

Construct	Analysis	Power at $-c_{bm}$	Power at $+c_{bm}$	Difference / SE
CS, ± 0.20	S	0.363	0.367	0.2
CS, ± 0.20	ES	0.418	0.416	0.1
AR(1), ± 0.30	S	0.447	0.481	1.5
AR(1), ± 0.30	ES	0.368	0.391	1.1

The estimates are mirror images (for S in the AR(1) construct, $+0.155$ and -0.158). The compound-symmetry cells agree exactly. The AR(1) cells again lean toward $+c_{bm}$, but within chance. Since the identity is exact, no amount of leaning can make it real.

Why the screen was a design error. It tested, with noisy and correlated comparisons, a property that holds by construction.

- **The false-failure rate was high.** Each test had about a 5% chance of false failure on its own. Because one simulated data set served all analyses, a single unlucky draw failed several analyses together.
- **The screen could only remove good analyses by chance.** It could not detect any real insensitivity, because none can exist in these cells.

There are two correct designs. Either state the symmetry as a property and test only one sign, or generate the $-c_{bm}$ trials from the same random draws as the $+c_{bm}$ trials with the biomarker reflected. The second makes any asymmetry in the code visible and asymmetry from sampling impossible.

Effect on the selection.

Table 10: Survivors and selected analysis with and without the symmetry screen

Rule	Survivors	Selected
Original protocol	C2, F1	C2 by maximin, but flagged by the prognostic screen; only F1 passes every screen
Symmetry screen dropped (a protocol revision)	S, C1, C2, DS, ES, EA, F1, F2	C1 by maximin (0.034), but flagged; among the unflagged, ES on non-prognostic cells, F1 on all cells

Either way, the analyses that are both nominal and robust to a prognostic biomarker are ES, EA and F1, and the recommendation of F1 does not depend on the faulty screen. `docs/47` will present the outcome under the original protocol first and the revision second.

With the three post hoc analyses added (2026-10-07), the survivors without the symmetry screen also include G1 and G2, the robust set is ES, EA, F1, G1 and G2, and F1 is selected on the tie rule (`docs/46`, Amendment 2).

5. Combining F1 with phase-specific variances (tested post hoc)

The idea. C1 and C2 were the most powerful analyses in nearly every non-prognostic cell. For example, at $c_{bm} = 0.30$ in the AR(1) construct, C2 reached 0.606 and C1 0.578, against 0.514 for S and 0.364 for F1. F1 is the most robust. The combination is C1 with F1's terms:

```
nlme::lme(Sx ~ bmc + t + Dbc + bmc:Dbc + De + bmc:De,
          random = ~ 1 | ptID,
          correlation = nlme::corCAR1(form = ~ t | ptID),
          weights = nlme::varIdent(form = ~ 1 | phase))
# CR2 standard errors as in S
```

Why it might combine the two strengths (inferred):

- **The power of C1** probably comes from weighting. The implied variance differs by phase (`docs/40`: baseline 342, open label 710, blinded 599 in the published construct). A model that lets the variance differ weights each visit by its actual precision.
- **The robustness of F1** comes from its mean model. `bmc:De` frees the baseline slope and absorbs an expectancy-scaled prognostic slope.
- **The two operate on different parts of the model,** variance and mean, so in principle neither undoes the other.

Why it might not:

- **The weighting that gives C1 its power is the weighting that made it fail worst under a prognostic biomarker** (0.382 against 0.099 for S). It up-weights the baseline row. If F1's terms free the baseline slope, that extra weight should become harmless, because it now feeds only the baseline-specific slope, not the interaction. But this is the untested crux.
- **Freeing the baseline slope removes the baseline information** that helped the shared-slope analyses. The power gain from phase variances may shrink once baseline no longer informs β_B. In the non-prognostic cells, the phase variances might then add little over F1 alone.
- **More variance parameters** (two more) may cost something at $N = 35$, where C1 was already the highest of the CR2 analyses (0.069, within the cell limit).

Status (as drafted). Untested. It was not among the analyses of the original protocol, so any result is post hoc. It can motivate a recommendation, but it cannot replace the protocol's selection without an independent run.

The test. One arm on the same 39 cells and the same trials, run as a post hoc extension of `11-strawman-stress-test.R`, takes about 3 to 4 hours. It succeeds if two conditions hold:

- it is nominal in all 11 standard null cells and in all three prognostic cells;
- its size-adjusted power exceeds F1's by a margin worth its complexity, say 0.05 in the AR(1) cells.

Result (2026-10-07; verified, `docs/46`, Amendment 2). The combination was fitted with CR2 inference (G1) and with model-based inference (G2), on all 41 cells.

Table 11: Size and size-adjusted power of F1 and its phase-variance versions G1 and G2

Analysis	Pooled null, 11 standard cells	Prognostic null cells (4)	AR(1), $c_{bm} = 0.30$	CS, $c_{bm} = 0.25$	Maximin shortfall, non-prognostic
F1	0.051	0.040 to 0.047	0.364	0.522	0.242
G1	0.049	0.042 to 0.055	0.354	0.546	0.252
G2	0.047	0.043 to 0.062	0.380	0.528	0.232

- **The first condition holds.** Both versions are nominal in every standard and prognostic null cell, the baseline-severity cell included. The extra weight on the baseline row is harmless once the baseline slope is freed, as hoped.
- **The second does not.** Averaged over the 16 non-prognostic alternative cells of the AR(1) construct, the gain over F1 is 0.002 for G1 and 0.017 for G2; the largest single-cell gain is 0.070 (G2, Weibull $k = 0.5$). At $c_{bm} = 0.30$ it is -0.010 and $+0.016$ (verified). The power C1 had over S does not survive freeing the baseline slope, which supports the second reason for doubt above.
- **Selection.** G2 and F1 tie within 0.02 on maximin shortfall; neither reaches power 0.80 on the grid, so the tie goes to F1, which has fewer covariance parameters. Over all alternative cells, prognostic ones included, G1 (0.308) and G2 (0.312) lead F1 (0.358), mainly through the natural-course prognostic alternatives. Neither reading justifies replacing F1 with a post hoc analysis.

6. Summary

- The shared-slope analyses, the strawman, the unstructured model and the published analysis among them, gain power by assuming the biomarker's off-drug association is the same at baseline and afterwards.
- A prognostic biomarker violates that assumption and produces false interactions. These reach 0.10 for the strawman and 0.42 for the unstructured model.
- Freeing the baseline slope (ES, or F1 through the expectancy term) removes the problem. Under AR(1) data it costs about a third of the power; under compound symmetry it costs little.
- The symmetry screen was a design error: it tested an exact identity with noisy comparisons. Dropping it does not change which analyses are both nominal and robust.

- Phase-specific variances combined with F1's terms (G1, G2) hold their size everywhere but add no material power in the AR(1) cells; F1 remains the recommended analysis (updated 2026-10-07; `docs/46`, Amendment 2).

7. Reproducibility

```
Rscript analysis/scripts/quick-sim/carryover-closed-form/11-strawman-stress-test.R \
  --cores 8 --chunk 25
Rscript analysis/scripts/quick-sim/carryover-closed-form/12-strawman-stress-decision.R
Rscript analysis/scripts/quick-sim/carryover-closed-form/13-symmetry-check.R \
  --reps 1000 --cores 8
# post hoc analyses (G1, G2; F1A) and the 14-analysis decision
Rscript analysis/scripts/quick-sim/carryover-closed-form/11-strawman-stress-test.R \
  --cores 8 --chunk 25 --extra
Rscript analysis/scripts/quick-sim/carryover-closed-form/11-strawman-stress-test.R \
  --cores 8 --chunk 25 --extra2
Rscript analysis/scripts/quick-sim/carryover-closed-form/12-strawman-stress-decision.R \
  --strawman DS --no-symmetry --with-extras
```

Outputs are in `analysis/data/quick-sim/carryover-closed-form/strawman-stress/`:

- `summary-full.csv` and `replicates-full.rds`;
- the `decision-*.csv` tables;
- `symmetry/summary-reps1000.csv`.

8. References

1. pmsimstats team. `docs/40-hendrickson-hybrid-design-primer.md`, `docs/41-baseline-as-covariate.md`, `docs/45-correctly-sized-hybrid-analysis.md`, `docs/46-strawman-stress-test-protocol.md`.
2. Liu GF, Lu K, Mogg R, Mallick M, Mehrotra DV. Should baseline be a covariate or dependent variable in analyses of change from baseline in clinical trials? *Statistics in Medicine* 2009; 28(20):2509-2530. doi:10.1002/sim.3639.
3. Bell RM, McCaffrey DF. Bias reduction in standard errors for linear regression with multi-stage samples. *Survey Methodology* 2002; 28(2):169-181.